

Breakdown radius for background knowledge

What we have measured, and the three places it changes the plan

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Where we agree, which is most of it

Your rewrite settles four things that were open since 14 August, and settles them the right way.

- **The object.** A breakdown radius, $r_{\text{opt}} \leq r_{\text{val}}$ kept apart, and §5 says the radius is not a probability. The “bound” line is closed.
- **The reference.** Taeb, Guo & Henckel is the methodological pivot, and §10 is right that the version must be pinned. It is **v3, 2026-06-30**.
- **The positioning.** Metricity demoted to a corollary (§6); §10 refuses to lead with their open problem.
- **The literature you bring.** Masten & Poirier on breakdown frontiers, Horowitz & Manski. Neither appeared anywhere in our corpus before your document.

The one thing to settle first

Your message says “très empirique avec des valeurs nominales”. Your §10 recommends UAI or AISTATS and the document carries no measured number. Nothing else decides until this does.

Three things that change the plan, and two that are already done

1. **The unit.** A flip costs two, so every number we hold moves by a factor of two.
2. **Locality is the result**, not a pruning heuristic in §8.2.
3. **§2 puts b-LOAD outside the framework**, and not cleanly.
4. **WP0(d), the decisive statistic**: measured, three times, two runs.
5. **The §4.3 calibration experiment**: ran on 19 August.

Plus one scope condition we have both been leaving implicit: **amenability**.

1. The unit changes every number we hold

§4.1 prices a flip at two: a retraction followed by an assertion.

Our pilot's operator *is* a flip. It reverses an orientation that exists in the CPDAG skeleton, and the budget is global at $\text{MAX_K} = 4$.

	our unit	your unit
one expert error	1 (a flip)	2
pilot budget $\text{MAX_K} = 4$	4	8
$\rho^* = 1$, breaks at the first error	1	2

It cuts both ways. Our feasibility evidence is **stronger** than §8.2 assumes (radius 8 on $p \leq 10$, twenty minutes of CPU). And every censoring figure below sits on a **different abscissa** from the one your figures will use.

2. Locality is the finding, not a speed-up

§8.2 lists “prune by relevance” among the computational mitigations. That quotient is the result.

Censoring of ρ_{any}^* at $n = 20\,000$, by minimum hop distance from an elicited statement to $\{X, Y\}$ on the CPDAG skeleton:

ensemble	dist 0	dist 1	dist ≥ 2
original (p 5–8)	0.037 _(2 539)	1.000 ₍₉₃₎	
licensed (p 5–10)	0.070 _(1 852)	0.935 ₍₂₂₉₎	
k8, $ K = 8$	0.016 ₍₅₆₃₎	0.955 ₍₂₂₎	
large (p 15–25)	0.330 _(1 472)	0.977 ₍₆₃₈₎	1.000 ₍₃₃₁₎

13 to 27 fold. Invariant in n to four decimals (0.9345 at $n = 2\,000, 20\,000, \infty$). Exactly 1.000 for all 331 large SCMs two or more hops out.

ρ^* is a functional of O^* and O^* is local, so a statement that does not reach the query cannot move it. No sample size repairs that.

2b. What locality buys, and one anomaly it dissolves

The second currency the abstract is missing

Elicited knowledge far from the query does not move the estimate. **Stop paying to elicit it.** A cost that goes down, with a number, is one of the two claims that survive adjustment for domain.

It also dissolves something we were both treating as a separate regime. large behaves differently because only **60.3 %** of its SCMs have a statement at distance 0, against 88.7 % (licensed) and 96.3 % (original).

Condition on distance 0 and it stops being a different regime.

Your 22 August note reaches the same place from the other side: if influence is local, one false statement touches $O(1)$ of $O(d^2)$ queries, so mean damage tends to zero as the graph grows. Aggregate worst-case or distributionally, never on the mean.

3. The §10 risk is real at both ends

§10 fears $r_{\text{val}}(O) = 1$ almost always. Both extremes are true at once in our runs.

Saturates at the top

Censoring of ρ_{any}^* :

0.554 at $n = 200 \rightarrow$ **0.515** at $n = \infty$.

3.9 points over *five orders of magnitude*,
against a hard floor.

Because **64.6 %** of problems have $d(1) = 0$
exactly, at every n . The ball is *inert*, not
fragile.

Saturates at the bottom

Repair the other end with the composite

$\min(\rho_{\text{se}}^*, \rho_{\text{ident}}^*)$:

censoring falls to **0.166**, and **69.0 %** of the
mass lands at $\rho^* = 1$.

That is literally your principal risk.

The switching variable is **hop distance**, and it is not in the document.

3b. So WP0 has one answer per stratum

Your three branches are not alternatives. They are strata.

§10 branch	when it holds
the O-set always breaks at 1	near the query
the O-set survives	away from it
binary validity is the wrong criterion	where the ball is inert, i.e. most of the mass

An unstratified distribution of r_{val} averages two regimes and describes neither.

One consequence for the venue paragraph

The empirical ICLR hook you propose, *knowledge-informed methods break at radius 1–2 on standard benchmarks*, is **contradicted by our data as written**: more than half of problems never break inside the budget. The **conditional** version survives, and is stronger, because it tells the practitioner what to do.

4. §2 puts b-LOAD outside the framework

§2: adjacency knowledge is vacuous or refuted, so the space is forced to be orientations. The argument is good. The consequence is uncomfortable.

b-LOAD's noise operator asserts a required edge on pairs that are mostly **non-adjacent**. Under §2 those are refuted, so they fall outside $\mathfrak{G}_c$ and outside the theory.

We measured that disjointness on 14 August. §2 does not repair it. It formalises it.

That matters because the empirical demand driving this whole line sits on that side: all four CIKM reviewers and the metareview raised the perfect-knowledge assumption, and R4 asked specifically for a robustness analysis on Intervention Distance *in the main body*.

4b. And the exclusion is not clean

b-LOAD's own sampler, on licensed (8 000 draws):

class	count	rate
pure_spurious (outside the space)	5 555	69.4 %
reversal_of_true_edge (inside it)	2 445	30.6 %
the query pair itself	1 030	12.9 %

original 73.1 / 27.0, large 91.4 / 8.6. Quote the ensemble you benchmark on. **Oracle labels: no rate stratified by this is a triage rule.**

The way out is yours, and it is publishable

§2 offers a *free consistency check*. Applied to b-LOAD it catches **all** of its spurious noise, and answers R4 directly. Written honestly it reverses the story: b-LOAD's robustness stops being a surprise and becomes a check that was never run. **The current code cannot perform it: the assertion is written into the initialisation matrix and then protected.** Decide which of the two papers eats that sentence.

5. Amenability needs to be a fourth exit beside $\perp$

$\text{cl}(\mathcal{G}, a \rightarrow b)$ returns $\perp$ for a cycle or a forbidden structure. It says nothing about $O(\mathcal{G})$ being *undefined* because the MPDAG is not amenable.

The fate of one wrong orientation, 2 058 perturbations:

outcome	of all	of Meek-consistent
Meek-inconsistent, caught free	33.0 %	—
O^* unchanged, zero bias	42.3 %	63.1 %
not amenable, <i>loud</i> failure	15.3 %	22.9 %
O^* changed and invalid, <i>silent</i> bias	9.4 %	14.1 %
O^* changed but still valid	0.0 % (0/754)	0.0 %

A loud failure is a different event: the analyst sees it. Pricing it as a breakdown inflates fragility, the objection §4.2 already makes against collapsing r_{opt} and r_{val} .

And note the last row: the efficiency-only band THEORY.md calls “the one worth naming” measured **empty** on our operator.

6. WP0(d) is done, and §3 wins its argument

§3: the case against naive assertion counting turns on the fraction of single flips that are logically inconsistent with $\hat{C}$. If it is substantial, the naive baseline is not coarser, it is incoherent.

It is substantial. Three measurements, two independent runs.

Meek-inconsistent, all single errors (pilot, 17 July)	33.0 %
Meek-inconsistency rate, G1 denominator (E1', 19 Aug)	32.4 %
caught free by Meek, ARM 2 single errors	23.6 %
caught free by Meek, ARM 2 double errors	38.6 %

The pilot and E1' agree to within half a point, from separate code paths.

§3 asks for this to be “demonstrated, not argued”. It already is.

7. The §4.3 calibration experiment ran on 19 August

“Corrupt the knowledge, compute r around the corrupted $\mathcal{G}_0$, check whether $d(\mathcal{G}_0, \mathcal{G}_{\text{true}}) < r$ predicted validity.”

That is ARM 2 of E1': **2 500 SCMs** $\times$ 20 datasets. Coverage at $r = 0$ is **0.9509** against 0.95 nominal, zero regeneration-gate failures.

And it produced the number §5 asks for, the one comparable to a standard error ($n = 20\,000$):

	$r = 0$	$r = 1$	$r = 2$
naive CI coverage	0.951	0.720 [0.713, 0.727]	0.568 [0.559, 0.577]

One reversed statement costs **23 points** of coverage. Two cost **38**.

Among the errors Meek does not catch for free.

7b. And the ignorance does not shrink with data

$SE_n \propto n^{-1/2}$, while $d(\rho)$ is free of n . So the two do not converge, they diverge.

n	200	1 000	5 000	20 000
mean robust / naive width ratio	1.61	2.32	3.93	6.84

This is the best answer we have to “collect more data”. Structural uncertainty about the orientations is not a small-sample problem that goes away, and at $n = 20\,000$ it dominates the sampling noise by a factor of seven on the mean.

With the caveat from the next slide: the *median* ratio is 1.00, because the ball is inert for 69 % of problems. Print both.

Two numbers of ours you should stop using

I would rather you heard these from me than found them.

0/791 is dead

Our own audit on 19 August marked it *no provenance*. It is the figure your 22 August assessment quoted back at us. The locality claim is carried by the censoring table, which is a frequency, not a zero-existence result.

The median width ratio of 1.00 flatters the method

The ball is inert for 69 % of problems, so the median is 1.00 by arithmetic. I pre-registered the median. The mean is **6.84** $\times$, which would have failed the pre-registered $3\times$ bar. Both belong in the paper.

What I need from you before 18 September

1. **Read Taeb, Guo & Henckel v3 in full and tell me what §6 actually does.** Our corpus note is explicit that everything it says about §6, Lemma S6 and the open covering relation is second-hand from your framing. **If their adjustment-set application already defines a tolerance radius centred on the analyst's graph, the novelty surface shrinks a lot.** I would rather know now than in a review.
2. **The covering enumeration on four and five vertices** (your WP2). It doubles as the independent correctness check on a preprint no referee has read, which §10 correctly treats as a small credibility gain.
3. **Guo & Perković 2010.08611 in full.** The framing rests on “they never perturb K ” and nobody on either side has read it.

From me

The code, on branch `rho-breakdown-evidence`. It is what you asked for on 21 August, and I am sorry it took ten days.

What is on the branch

pilot/	17 July: 600 + 1 650 + 60 SCMs, iSCM, closed-form linear SEMs
e1prime/	19 Aug: the se criterion, the censoring floor, ARM 2
locality/x1	hop-distance stratification, operator arms, pool composition
locality/x2	exhaustive enumeration, no SCM and no truth, complete at $p \leq 5$
REPORT.md	this deck, in prose, point by point
NUMBERS.md	every figure mapped to the file it came from

Each run directory carries a PREREG.md written before the first number and an AUDIT.md written against the design.

CPU only. numpy, scipy, pandas, networkx. No GPU, no private data.

NUMBERS.md also lists the four corrections we applied to ourselves, and the three things never run: an estimated CPDAG, nonlinear mechanisms beyond the pilot's 60, and the skeleton perturbation that would repair the oracle-CI assumption in your §2.